DOI: 10.1111/emre.70042

RESEARCH ARTICLE

EMR European Management Review The Journal of the European Academy of Management

Managing artificial intelligence across functions for enhanced retail firm performance

Lanlan Cao[1] | **Jiachen Yang**[2] | **Yen-Tsang Chen**[3]

[1]Marketing at NEOMA Business School, Mont-Saint-Aignan, France

[2]Strategy & Entrepreneurship at NEOMA Business School, Mont-Saint-Aignan, France

[3]Supply Chain Management at NEOMA Business School, Reims, France

Correspondence
Lanlan Cao, Marketing at NEOMA Business School, Boulevard André Siegfried, 76825 Mont-Saint-Aignan, France.
Email: lanlan.cao@neoma-bs.fr

Abstract

Firms are increasingly deploying AI across business functions, yet the core challenge lies not in adoption itself, but in integrating these applications to achieve coherent, firm-level outcomes. Despite growing interest, prior research has largely overlooked the dynamic interdependencies, unpredictable interactions, and organization-wide effects that shape how AI-enabled functions collectively drive performance. Addressing this gap, we draw on a business process perspective and dynamic capability theory to develop a multi-level framework of AI integration at the task, function, and firm levels. Using a sequential mixed-methods design, we first construct a validated measurement instrument based on grounded analysis of 6,519 AI-related documents from 37 retailers. We then apply fuzzy-set qualitative comparative analysis (fsQCA) to survey data from 140 executives to identify configurations of AI-enabled functions associated with efficiency, innovation, or both. The results show that superior performance stems not from isolated AI uses but from synergistic combinations—particularly those involving customer service and cybersecurity—interacting with other functions. These configurations appear to give rise to emergent capabilities such as adaptive learning, predictive analytics, and uncertainty mitigation, enabling firms to reconcile exploitation and exploration. This study offers a dynamic, process-based view of AI capability and provides strategic guidance for designing ambidextrous AI portfolios in retail.

KEYWORDS

AI implementation, artificial intelligence, business process management, innovation configuration, organizational efficiency, retail performance

INTRODUCTION

The AI market in retail is expanding rapidly, with investments projected to reach USD 826.73 billion by 2030, reflecting substantial growth (Thormundsson, 2024). This surge has intensified interest in understanding AI's impact on organizations, particularly its role in business value creation and firm performance (Ardito et al., 2025; Fosso Wamba et al., 2024; Mikalef & Gupta, 2021; Sullivan & Fosso Wamba, 2024).

Early research conceptualized AI as a set of discrete technological elements that influence specific organizational outcomes (Li et al., 2021; Mishra et al., 2022; Wang et al., 2023). However, recent studies emphasize AI's dynamic and interactive role, embedding it within complex business processes rather than treating it as an isolated technology (Berente et al., 2021; Bosse et al., 2023; Kemp, 2024). This shift underscores the necessity of a systematic approach to assessing AI's interactions within business contexts to accurately evaluate its impact on firm performance.

Mikalef & Gupta (2021) propose AI capability as a measure of AI's interactions within organizations. However, their framework lacks contextual specificity, overlooking AI's diverse applications across operational settings. Growing research (e.g., Kemp, 2024; Kopalle et al., 2022) highlights the importance of context in allowing AI to overcome strategic limitations, such as its generic nature, reliance on explicit knowledge, and tendency toward myopic applications. Contextualizing AI within specific business environments enables firms to leverage AI more effectively, fostering competitive advantages and improving performance.

To address this gap, scholars advocate for operationalizing AI capability through AI-enabled business processes, offering a context-sensitive framework for assessing AI's organizational impact (Bosse et al., 2023; Heller et al., 2021; Manis & Madhavaram, 2023). However, existing assessment tools primarily provide broad overviews of AI-enabled business processes, often failing to capture contextual nuances (Abadie et al., 2023), or they focus on limited aspects of AI implementation (Abou-Foul et al., 2023; Fosso Wamba, 2022). This underscores the need for a more comprehensive and context-aware measurement approach to AI-enabled business processes.

AI-enabled business processes are characterized by task-level human-machine interactions (Huang & Rust, 2022; Raisch & Krakowski, 2021) that integrate into broader business functions (Manis & Madhavaram, 2023), ultimately shaping firms' strategic objectives (Davenport et al., 2020). While prior research has predominantly examined AI's role within individual business functions – such as customer service, store operations, logistics, marketing, and cybersecurity (Ameen et al., 2021; Baryannis et al., 2019; Kumar et al., 2019; Shankar, 2018) – recent studies emphasize the importance of cross-functional AI coordination to fully leverage AI's potential and enhance firm performance (Abou-Foul et al., 2023; Božič & Dimovski, 2019; Mikalef & Gupta, 2021).

Although empirical studies (Abadie et al., 2023; Fosso Wamba, 2022) support the view that coordinating AI-enabled business functions is essential for performance improvement, they do not fully explain how these functions interact to generate organization-wide AI capabilities. This gap underscores the need to examine how different AI-enabled business functions combine and configure to drive firm performance.

To address these challenges, this study explores AI-enabled business functions and the complexity of their combined effects on firm performance, with a specific focus on efficiency and innovation as key performance criteria. These dimensions are widely recognized as core mechanisms explaining AI's impact on firm performance (Akter et al., 2023; Babina et al., 2024; Sullivan & Fosso Wamba, 2024).

This study seeks to answer the following research questions:

1. How can AI-enabled business functions be conceptualized and systematically measured based on AI-enabled tasks?
2. What configurations of AI-enabled business functions drive firm performance in terms of efficiency and innovation?

To address our research questions, we adopt a business process perspective (Zammuto et al., 2007) and dynamic capability theory (Teece et al., 1997) to develop a framework categorizing AI-enabled processes at the task, function, and firm levels. This framework systematically analyzes their interactions, forming the basis of our research model. We employ a sequential mixed-method approach: Study 1 uses grounded theory to analyze AI applications from 37 retailers, identifying 28 AI-enabled tasks across five business functions and developing a measurement tool for task-level AI assessment. Study 2 applies this tool in a survey of 140 retail executives, using fsQCA to identify AI configurations driving efficiency and innovation.

Our study contributes to the literature by (1) developing a structured framework to assess AI-enabled business functions through task-level applications, (2) identifying AI configurations that enhance efficiency and innovation, (3) revealing interdependencies among AI functions, and (4) offering actionable insights for optimizing AI implementation in retail.

THEORETICAL BACKGROUND AND FRAMEWORK

AI Technologies in Organizations

While there is no universally agreed-upon definition of AI, it generally refers to technologies capable of mimicking human intelligence (Bosse et al., 2023; Shankar, 2018). AI technologies encompass a wide range of programs, algorithms, systems, and machines, such as machine learning (ML), deep learning, natural language processing (NLP), computer vision, robotics, expert systems, speech recognition, reinforcement learning, and generative adversarial networks (GANs) (Kopalle et al., 2022). AI is distinct from other technologies in that it can learn from data and adapt autonomously over time, reshaping human actions and interactions (Huang & Rust, 2022; Kopalle et al., 2022). Implementing AI involves integrating AI features with organizational structures and practices that support their applications (Berente et al., 2021).

Studies of new and emerging technologies have been central to organization theory for some time. The existing literature on technology in organizations is diverse (Bailey et al., 2019). Orlikowski & Scott (2008) categorize this literature into two research streams based on their perspectives on the nature of technology, its role in organizations, argument logic, key mechanisms, and methodological orientation.

AI as discrete entities

The first research stream, exemplified by works such as Attewell & Rule (1984), Huber (1990), and Dewett & Jones (2001), conceptualizes technology as discrete entities interacting with different facets of an organization. Here, technology is often regarded as an independent variable directly affecting various organizational outcomes. For example, Mishra et al. (2022) and Wang et al. (2023) adopt this viewpoint and examine AI technology's direct effect on firm performance like net profitability, net operating efficiency, return on marketing-related investment, and innovation. Technology is also considered a moderating variable that shapes the relationship between organizational factors and outcomes in this stream. For instance, Li et al. (2021) investigate the moderating effect of AI on the relationship between firms' corporate social responsibility and risk reduction.

This research stream operationalizes technology in several ways, including quantifying it by the number, type, or cost of machinery, devices, techniques, etc. For example, Wang et al. (2023) measure AI technology in a firm by the number of installations and the operational stock of robots. Studies in this category typically use a variance-based approach in their research design (Mohr, 1982).

AI as part of a complex process

The second strand of research (e.g., Barley, 1988; Roberts & Grabowski, 1996; Zammuto et al., 2007) views technology as an integral element in the complex processes that shape organizational activities. This perspective highlights the dynamic interactions between individuals (or organizations) and technology. Recent studies further illustrate this approach: Bosse et al. (2023) and Bonetti et al. (2023) explore the interplay between AI and stakeholders, emphasizing the effects of AI deployment on firm performance. Zirar et al. (2023) conducted a systematic review to examine value creation resulting from the coexistence of AI with workers and workplace environments. Similarly, Bankins et al. (2024) review empirical research to develop a multilevel framework that identifies individual, group, and organizational factors influencing human-AI interactions. Petrescu et al. (2024) illustrate this view by showing how AI technologies are embedded in consumer-retailer interactions along the customer journey. These interconnections are characterized as embedded and emergent, resisting deterministic interpretations (Meyer et al., 2005).

Significantly, Berente et al. (2021) emphasize AI's escalating complexity as it gains knowledge and autonomy, making its management distinct from traditional IT and requiring evolving capabilities. We adopt this perspective, framing technology outcomes as interdependent co-evolutions between AI and organizations over time—not treating technology as an independent variable.

Mikalef & Gupta (2021) define AI capability through three AI-specific resources as a second-order formative construct reflecting firm–AI interactions, but their framework lacks contextual grounding. Kopalle et al. (2022) and Kemp (2024) stress contextualization's critical role. Kemp's (2024) "situating AI" addresses three strategic limitations – AI's generic nature, reliance on explicit knowledge, and myopic applications – to enhance competitive advantage. However, none of these studies offered methods to operationalize AI capability contextually, underscoring the need for alternative approaches.

Building on this gap, the literature (Bosse et al., 2023; Heller et al., 2021) proposes operationalizing AI capability through the lens of business processes. This aligns with the second research strand, which advocates using the concept of affordance to examine how technology creates new capabilities by integrating into specific uses (e.g., business processes) (Zammuto et al., 2007). However, existing assessment tools often provide only a general overview of AI-enabled business processes, without addressing contextual specifics (Abadie et al., 2023), or they focus on narrow aspects of AI implementation (Abou-Foul et al., 2023; Fosso Wamba, 2022). This gap motivates us to further explore the concept of AI-enabled business processes to develop a relevant measurement framework, refine our positioning, and propose robust research models.

AI-enabled Business Processes

A business process can be defined as a set of interconnected subsystems – comprising people, structure (tasks, activities, and areas of responsibility), technology, and other elements – that work together to deliver value to customers or achieve strategic objectives (Guha et al., 1993; Strnadl, 2006; Melão & Pidd, 2000). AI-enabled business processes highlight the pivotal role of AI technology in enhancing or transforming specific organizational functions through its interactions with various subsystems (Bosse et al., 2023; Heller et al., 2021; Zammuto et al., 2007). For instance, Manis & Madhavaram (2023) demonstrate how AI-enabled processes can shape and strengthen a firm's marketing capabilities. Building on these insights, our study seeks to explore the broader impact of AI-enabled business processes, extending beyond marketing to include other affected business functions, with a particular focus on the interactions and synergies between these functions.

Viewing a business process as a complex, dynamic, and emergent system implies that it can be hierarchically broken down into further levels of subsystems (Melão & Pidd, 2000). We examine business processes at three levels (i.e., task, function, and firm level) and discuss the key issues of AI applications at each level.

AI applications at the task level

AI use at the task level represents the most foundational aspect of AI-enabled business processes. The primary interactions within these processes revolve around human–machine interaction (HMI). HMI encompasses various forms of communication and interaction between individuals and automated systems, such as touch, gestures, voice, and sensors (Kopalle et al., 2022). AI exhibits relative strengths in mechanical and analytical intelligence, while human intelligence (HI) excels in contextual, intuitive, and emotional intelligence. Given these distinct strengths, AI–HI relationships may involve augmenting or replacing human labor (Huang & Rust, 2022).

Raisch & Krakowski (2021) define augmentation as scenarios where humans collaborate closely with machines to perform tasks, whereas automation refers to situations where machines take over tasks previously performed by humans. For instance, senior managers at Stitch Fix report that their AI is most effective when it augments human stylists' capabilities in selecting suitable clothing for each customer. By contrast, Kanetix automated its customer targeting task by implementing a machine learning model, yielding a 230% return on its AI investment (Davenport et al., 2020).

A task typically focuses on a specific level of intelligence – either augmentation or automation (Huang & Rust, 2022). Recognizing the foundational nature of AI applications at the task level, Davenport et al. (2020) recommend considering both intelligence levels and task types to understand the impact of AI. The alignment of these dimensions determines outcomes, specifically how business processes achieve goals (Garvey et al., 2023; Huang & Rust, 2022). Therefore, we propose a task-level analytical framework for AI in retail, encompassing three elements: intelligence level (augmentation or automation), specific task, and desired objectives.

AI applications at the function level

The second level, AI uses in business functions, involves a set of AI-enabled tasks related to the same area of responsibilities within an organization to achieve specific goals (Huang & Rust, 2022, Melao & Pidd 2000). The primary business functions where AI is applied vary across industries (Garbuio & Lin, 2019). In this study, focusing on the retail sector, we adopt Cao's (2021), Guha et al.'s (2021), and Shankar's (2018) framework to classify AI-enabled tasks into five main blocks: customer service, store operations, logistics and supply chain, marketing, and cybersecurity and risk management.

Customer service
Firms use AI, primarily through self-service technologies, to transform their interactions with customers at various stages of the customer journey (Ameen et al., 2021; Heller et al., 2021; Hollebeek et al., 2021).

Store operations
Firms utilize AI technologies independently or in conjunction with store staff to enhance operational efficiency and improve the shopping experience (Balakrishnan & Dwivedi, 2024; Shankar, 2018).

Logistics & Supply Chain
AI enhances supply chain efficiency and optimizes inventory management and logistics (Mahroof, 2019; Shankar, 2018).

Marketing
AI technology operates in automation and continuous learning, serving as the intelligence behind data-focused analytics and decision-making (Davenport et al., 2020; Kumar et al., 2019; Manis & Madhavaram, 2023).

Cybersecurity and risk management
AI autonomously determines actions to help companies protect against cyber threats and achieve other risk management objectives (Akter et al., 2022; Baryannis et al., 2019).

AI applications at the firm level

At the firm level, the primary challenge of AI implementation lies in integrating applications across business functions to achieve strategic outcomes (Fosso Wamba, 2022; Mikalef & Gupta, 2021). This integration depends on dynamic capabilities – the firm's ability to coordinate and integrate resources to generate synergistic value beyond the sum of individual components (Teece et al., 1997). Building on the business process perspective and dynamic capability theory (Teece et al., 1997; Zammuto et al., 2007), effective AI management at this level involves selecting and orchestrating function-level applications to align with firm-wide objectives (Mikalef & Gupta, 2021).

Selecting and assessing AI-enabled business functions presents challenges due to the dynamic, emergent nature of the system, driven by interactions among various AI-enabled functions (e.g., Berente et al., 2021; Melão & Pidd, 2000). Our argument underscores the complexity of three dimensions:

Interdependence and Coordinated Change. AI-enabled business functions are interdependent and often change together (Meyer et al., 2005). Overlooking their interdependence limits the development of end-to-end solutions, hindering the realization of business value (Chui & Malhotra, 2018).

Unpredictable Interactions. Interactions between AI-enabled functions can result in unforeseen effects. For instance, Stitch Fix's "Style Shuffle" AI application unexpectedly matched stylists with specific customers (an unexpected benefit in customer service), in addition to its anticipated role in informing stylists about customer preferences (expected in the supply chain) but also (Davenport et al., 2020).

Organization-Wide Strengthening. Implementing AI strengthens organization-wide abilities, enhancing coordination and integration among functions (Bankins et al., 2024; Mikalef & Gupta, 2021). For example, AI applications can facilitate cross-functional teamwork by consolidating data and producing user-friendly visualizations (Akter et al., 2023). However, there is a lack of clarity in the literature about which AI-enabled functions contribute most effectively to strengthening the entire firm's capabilities.

Research by Meyer and colleagues (Meyer et al., 1993, 2005) emphasizes that understanding the effects of these numerous interactions requires methods beyond conventional research approaches. Surprisingly, this aspect has received limited attention in the literature concerning the impact of AI on firm performance. Studies like those by Fosso Wamba (2022) and Abadie et al. (2023) still use variance-based approaches and symmetric statistical techniques like structural equation modeling to examine the integrated impact of AI-enabled business processes across various functions on firm performance. To address this research gap, we follow the recommendation of Park & Mithas (2020) by adopting the configuration and fsQCA approach in this study. This approach will enable us to establish robust, data-driven configurations of AI-enabled business functions that enhance firm performance.

Research model

We view AI as integral to complex organizational processes, adopting business process (Zammuto et al., 2007) and dynamic capability perspectives (Teece et al., 1997) to examine its integration at task, function, and firm levels (see Figure 1). To advance understanding of how AI-enabled business functions combine to enhance firm performance, we examine their configurational effects on two key outcomes: efficiency and innovation.

Efficiency and innovation are prioritized as key performance criteria for three reasons. First, they are well-established measures of business process management quality (Benner & Tushman, 2002; Melão & Pidd, 2000; Trkman, 2010). Second, they are central mechanisms explaining AI's impact on firm performance. For instance, Akter et al. (2023) emphasize AI-powered service innovation, while Babina et al. (2024) show AI enhances efficiency and innovation, improving operational forecasting and learning. Similarly, Sullivan & Fosso Wamba (2024) highlight AI's role in enabling product and process innovation, reducing experimentation uncertainty, and enhancing learning. Rammer et al.

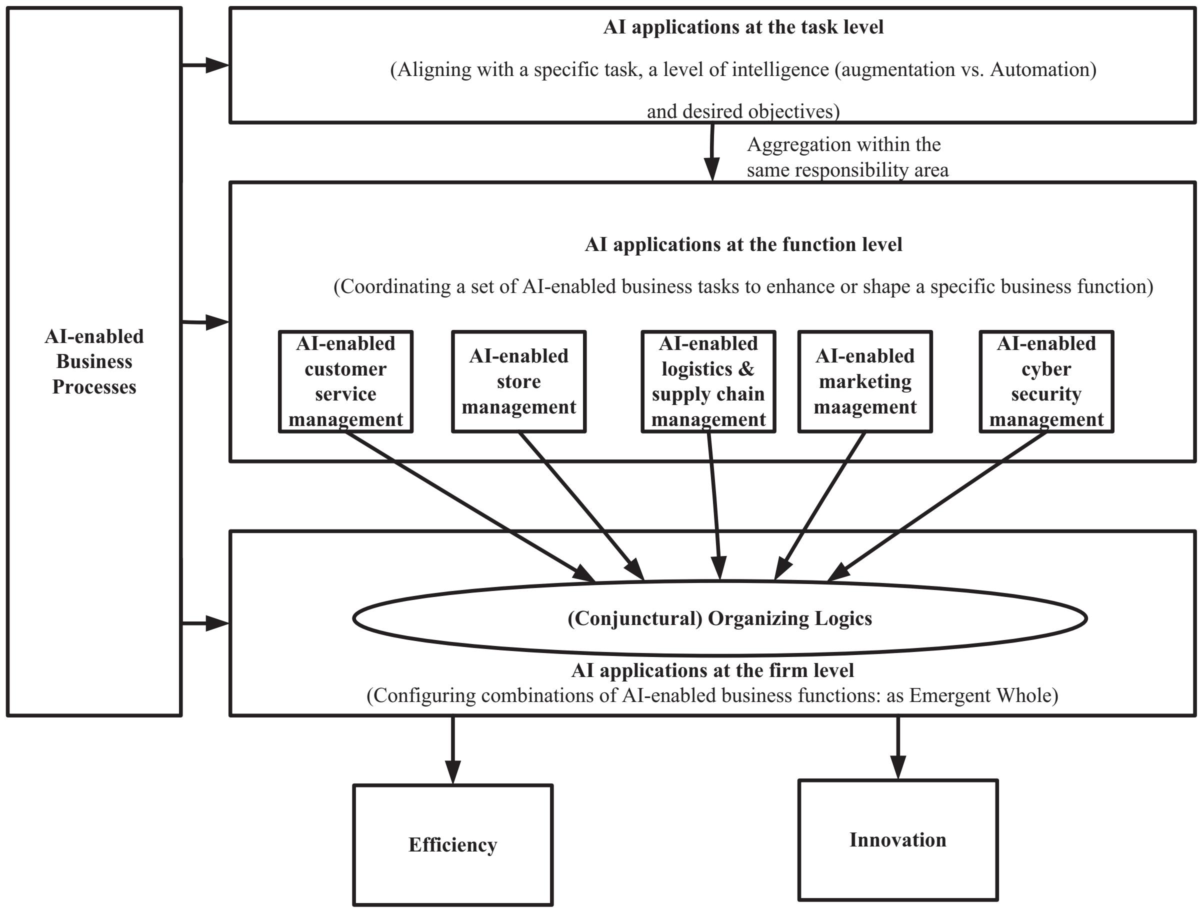

FIGURE 1 Research model.

(2022) provide large-scale evidence that AI adoption fosters both product and process innovation. Verganti et al. (2020) note AI's ability to integrate data and algorithms, boosting innovation speed and quality.

Third, these outcomes reflect two distinct but complementary innovation pathways – exploitation and exploration – in organizational learning theory (March, 1991; Raisch & Birkinshaw, 2008). In our model, efficiency represents exploitative innovation, achieved through AI-enabled cost reduction and process optimization, while innovation reflects exploratory efforts involving experimentation, personalization, and strategic renewal. This distinction provides a theoretical foundation for interpreting how different configurations of AI-enabled functions, when coordinated across domains, may contribute to exploitative or exploratory outcomes, or both.

RESEARCH DESIGN AND METHODOLOGY

We used a sequential mixed-method approach to develop and validate a measurement tool for AI-enabled business functions and explore their complementarities in driving firm performance. In Study 1, we applied grounded theory to analyze 6,519 Factiva articles (2017–2022) from 37 retail firms across diverse formats, categories, and geographies. Through iterative coding and expert validation, 28 AI-enabled tasks were classified into five business functions: customer service, store operations, logistics and supply chain, marketing, and cybersecurity and risk management. These tasks were operationalized as survey items for Study 2. In Study 2, we surveyed 140 U.S.-based retail managers to measure AI engagement and performance outcomes. FsQCA identified configurations of AI-enabled functions linked to superior performance. The sequential design enhanced methodological rigor, providing a structured understanding of AI's impact in retail.

CAPTURING AND CLASSIFYING AI-ENABLED BUSINESS PROCESSES AT THE TASK LEVEL INTO BUSINESS FUNCTION LEVEL (STUDY 1)

Prior research highlights the importance of operationalizing AI through AI-enabled business processes (Bosse et al., 2023; Heller et al., 2021), ideally starting at the task level (Davenport et al., 2020; Huang & Rust, 2022). However, as previously discussed, a comprehensive and context-sensitive approach to AI-enabled business process operationalization remains underdeveloped.

To address these gaps, we undertook an exploratory, inductive study to develop a more suitable measurement tool. This study employed grounded theory methodology (Corbin & Strauss, 2008). Our objective was to examine how retailers integrate AI technologies into their business processes. To this end, we sought to (1) identify AI-enabled tasks and (2) categorize them into five distinct business functions.

Research setting

We used the Factiva database, which covers a wide range of international media sources, to identify retail firms most frequently mentioned in the context of AI applications. Our search words were "artificial intelligence" and "AI", and we searched English-language articles in the retail industry published during 2017–2022. We selected the 2017–2022 period based on McKinsey's longitudinal studies (Chui et al., 2022) for three reasons. First, 2017 marked AI adoption's acceleration, with global usage doubling by 2022, capturing its shift from experimentation to broad integration. Second, advances in computer vision, NLP, and robotic process automation significantly boosted efficiency and innovation. Third, retail emerged as a leading sector in leveraging AI for customer experience, operations, and supply chains.

Ending in 2022 ensured mature AI applications for analysis and provided stable data for robust measurement. Our search yielded 6,519 articles, identifying 37 relevant retailers (Web Appendix A, Table A1) across 11 countries and five formats: specialty retail (56.76%), broadline retail (18.92%), consumer staples (10.81%), department stores (2.70%), and home improvement (10.81%).

Theoretical sampling

Using theoretical sampling (Corbin & Strauss, 2008), we selected retailers from 37 initial firms via Factiva to capture diverse AI contexts (Kopalle et al., 2022). We first analyzed 22 firms representing heterogeneous formats, products, and geographies. After adding 4 firms (total 26), no new concepts emerged – achieving initial saturation. Testing the remaining 11 firms revealed no new patterns, confirming the robustness of our findings.

Our sampling ensured variation across:

- Retail formats (e.g., specialty, broadline, department stores, home improvement retail).
- Product categories (e.g., food, apparel, drug, beauty, electronics).
- Geographic locations (retailers operating across different countries).

Data analysis

Utilizing the data from Factiva, we analyzed AI applications within each firm's task-level business processes, employing grounded theory content analysis and using NVivo 10 to facilitate this process. The analysis involved four steps (see Web Appendix A, Figure A1).

First, we assembled each firm's data. Second, using open coding (Corbin & Strauss, 2008), we identified preliminary concepts from the data, focusing on AI integration into task-level processes across three dimensions: intelligence level (augmentation/automation), specific tasks, and objectives (discussed in the theoretical background). An example (JD.com's checkout-free system) is in Web Appendix A (Figure A2). We then iteratively compared subsequent data units with prior ones, assigning identical or new labels as needed. This process yielded 39 empirical codes.

To enhance validity and reduce redundancy, two independent AI/retail experts, each with over ten years of experience in AI-driven retail operations, digital transformation, and strategy, validated the initial 39 empirical codes for conceptual clarity, relevance, and labeling accuracy. Overlapping codes were consolidated into 28 distinct AI-enabled tasks – e.g., merging "AI-based prevention from data loss" and "AI-based system to prevent firms from cyberattacks" into "AI-powered security platforms for automated threat detection" (Web Appendix A, Figure A3). This refined code definitions and ensured real-world alignment, strengthening the measurement tool.

In step three, through axial coding (Corbin & Strauss, 2008), we organized empirical codes into categories. Codes were grouped by responsibility area – e.g., *AI-powered semantic/image/voice search engines, recommendation algorithms, virtual try-on, voice command, and checkout-free systems* were categorized under "Customer service." This reflected two considerations: their direct relevance to customer service management and their collective role in transforming customer interactions across the journey.

Step four validated our results. We used systematic coding to capture task-level AI-enabled processes and classify them into business functions, following theoretical sampling and constant comparative methods until data saturation (Corbin & Strauss, 2008), which enhanced the interpretation reliability of our data.

To ensure validity and reduce bias, two reviewers evaluated a summary of our framework and definitions. Based on their feedback, we conducted axial coding, grouping 28 empirical codes into five categories (Table 1). These categories measured AI-enabled business processes across customer service, store operations, logistics and supply chain, marketing, and cybersecurity and risk management. We then applied the refined framework to the remaining 11 firms after analyzing the initial 26. The lack of new emerging concepts affirmed our measurement tool's effectiveness (Table 1; Corbin & Strauss, 2008).

To verify reliability, two experts – a food retailer's Chief Merchandising Officer and a digital imaging/AI firm's CEO – classified 28 items into our five categories (Web Appendix B). They achieved 25 out of 28 pairwise agreements (.89 intercoder agreement). Using Rust & Cooil's (1994) method accounting for categories and agreement, we calculated a proportional reduction in loss (PRL) reliability of .93, which exceeds Nunnally's benchmarks (.7 for exploratory; .9 for advanced research), confirming classification reliability.

Results and Discussion

Study 1 empirically validates the conceptual foundations proposed in prior research (Davenport et al., 2020; Huang & Rust, 2022), confirming that AI-enabled tasks serve as a critical framework for examining AI's interaction with organizations. Our study refines this framework across three dimensions: intelligence level (augmentation or automation), specific task, and intended objective. This structured approach enhances the precision of analyzing AI's role at the task level.

A key outcome of Study 1 is the development of a measurement tool (Table 1) to assess AI-enabled business functions. Using Factiva data on AI applications in 37 retail firms across various formats, categories, and geographies, we identified 28 distinct AI-enabled tasks. These tasks were classified into five core business functions: customer service, store operations, logistics and supply chain, marketing, and cybersecurity and risk management. This classification ensures that the measurement tool aligns with both theoretical frameworks and practical organizational processes, providing a systematic means to assess AI integration in retail.

Study 2 builds on this foundation by incorporating the measurement tool into a survey instrument (see Web Appendix C), assessing the extent of AI adoption in retail firms. The survey measured the frequency and scope of engagement with each AI-enabled task across business functions, ensuring consistency in constructs across both studies. This methodological alignment strengthens the coherence of our research design.

FSQCA ANALYSIS OF AI-ENABLED BUSINESS FUNCTIONS (STUDY 2)

FsQCA is a powerful analytical method increasingly applied in management studies (Fiss, 2007; Ragin, 2006), differing from traditional econometric methods like regression by identifying configurations of conditions that lead to specific outcomes (Fainshmidt et al., 2020; Pappas & Woodside, 2021). Unlike regression, which seeks a single solution, fsQCA identifies multiple sufficient or necessary solutions, known as equifinality. Studies by Abou-Foul et al. (2023), Mikalef & Gupta (2021), and others highlight AI's complex interactions across organizational levels, with tools working synergistically for cumulative effects (Guha et al., 2021; Shankar, 2018).

TABLE 1 Measurement tool for AI-enabled business functions.

Category (AI-enabled business functions)	Definition	Empirical codes (AI-enabled tasks)	Illustrative evidence
Customer Service	Firms use AI, primarily through self-service technologies, to enhance interactions across the customer journey.	AI-enabled semantic search engines for improved customer searches	Alibaba provides AI-powered semantic search APIs to enhance product discovery.
		AI-enabled image search engines for visual product search	Alibaba integrates AI-driven image recognition, allowing consumers to search for products using images.
		AI-enabled voice search engines to enhance search efficiency	1–800-Flowers enables customers to use voice commands via Amazon Alexa for product searches.
		AI-powered recommendation algorithms for personalized shopping	Amazon integrates AI-powered recommendations to enhance product discovery and user experience.
		Virtual try-on systems for personalized product selection	Gap Inc. deploys an AR-based app allowing customers to virtually try on clothing.
		AI-powered voice assistants for hands-free shopping and post-purchase services	1–800-Flowers enables customers to place orders using Amazon Alexa voice commands.
		Checkout-free systems for frictionless in-store purchasing	Amazon Go allows customers to shop without traditional checkout lines using AI-driven cashierless technology.
Store Operations	Firms deploy AI technologies to improve operational efficiency and enhance the in-store experience.	AI-powered visual merchandising for store optimization	Zmags' AI-driven ImageIQ analyzes customer preferences to optimize store displays.
		AI-powered category management for optimized product organization	eBay uses AI-driven "Grouped Listings" to categorize products dynamically.
		AI-powered merchandising algorithms for localized product assortment	JDA leverages AI to customize shelf layouts based on store-specific needs.
		Autonomous shelf-scanning robots for inventory tracking	Walmart employs autonomous shelf-scanning robots to monitor inventory levels.
		AI-powered atmospheric control for in-store environment optimization	Lawson Inc. uses AI-driven sensors to regulate store conditions for optimal customer experience.
		In-store chatbots to assist staff with customer service	Digital guides like E.L.F. help customers navigate large shopping centers.
		AI-powered sales assistance to support in-store staff	Aeon equips employees with IBM Watson-powered AI assistants to improve customer service.
Logistics & Supply Chain	AI enhances supply chain efficiency, inventory management, and logistics optimization.	AI-powered demand forecasting for optimized inventory replenishment	Myntra employs AI-driven demand forecasting to predict product demand.
		Autonomous order processing systems for real-time inventory control	Morrisons automates stock replenishment with AI-based optimization systems.
		AI-powered warehouse robotics for order fulfillment	Gap Inc. accelerates e-commerce fulfillment using autonomous robots.
		AI-driven sorting systems for efficient logistics	Flipkart implements AI-powered sorters in its fulfillment centers.
		Collaborative robots for warehouse automation	Ocado tests AI-powered warehouse robots to assist human workers.
		AI-powered drone delivery for last-mile logistics	Amazon develops PrimeAir drones to enhance delivery speed.
		AI-driven self-driving vehicles for automated delivery	Ocado introduces autonomous delivery vans for customer orders.
		AI-powered delivery modeling for route and logistics optimization	Scotts LawnService employs AI-driven logistics models to streamline service delivery.

TABLE 1 (Continued)

Category (AI-enabled business functions)	Definition	Empirical codes (AI-enabled tasks)	Illustrative evidence
Marketing	AI automates marketing processes, enabling data-driven decision-making.	AI-powered pricing algorithms for real-time dynamic pricing	Staples adjusts pricing on 30,000 products daily using AI-driven algorithms.
		AI-driven product catalog creation for automated content management	Akeneo employs AI to generate and enhance product descriptions.
		AI-powered ad spend optimization for targeted marketing	Tinyclues optimizes ad campaigns for Arcadia Group using AI-driven audience segmentation.
		Location intelligence for store site selection	Subway integrates AI-powered location intelligence to optimize franchise site selection.
Cybersecurity & Risk Management	AI autonomously strengthens cybersecurity and mitigates risks.	AI-powered security platforms for automated threat detection	Vectra provides AI-driven cybersecurity solutions to detect and prevent digital threats.
		AI-powered fraud detection to prevent online retail fraud	Amazon uses AI to combat fraudulent reviews and fake product ratings.

Despite growing adoption, the impact of AI-enabled functions on business performance remains underexplored. FsQCA's ability to identify AI-enabled function complementarities offers valuable insights for optimizing AI-driven innovation and efficiency.

Data collection

We used Qualtrics for data collection, employing theoretical sampling to target retail managers in U.S.-based companies with active AI investments (Thormundsson, 2024). Eligible respondents were involved in strategic or operational decision-making. We screened out ineligible participants, ensured confidentiality, and offered financial compensation to minimize bias. Respondents could withdraw at any time, and survey completion implied consent. Data were collected in a single wave over seven days, exceeding the quota, suggesting non-response bias is unlikely.

Before full data collection, we conducted a pilot test using two Qualtrics soft launches to refine the questionnaire. We gathered a total of 140 complete responses. Among these respondents, 25% were chief executives, while 75% were directors or managers at the business function level. The respondents indicated that their firms primarily operated in the following product categories: fast-moving consumer goods (34.3%), apparel (51.4%), hard goods (9.3%), general merchandise (7.9%), hospitality (2.1%), and others (0.7%).

Measure

AI-enabled business functions. This is a set of 5 formative variables derived from the results of study 1: AI in Customer Services (7 items), AI in Store Management (7 items), AI in Supply Chain Management (7 items), AI in Marketing Management (4 items) and AI in Cybersecurity Management (2 items). Our questionnaire assessed the engagement of their companies in each of the 28 AI-enabled tasks (as detailed in Table 1). Each task was rated using a seven-point scale, focusing on the "frequency of occurrence" rather than "importance," following the recommendations of Moncrief (1986). Participants rated their responses from 1 to 7.

Efficiency and Innovation. For organizational performance, we chose to measure AI value creation rather than traditional financial performance to mitigate potential endogeneity. Drawing on the work of Sorescu et al. (2011) and Zott & Amit (2007), we measured efficiency using four items: "save on labor costs," "save transaction costs," "increase the speed of the value chain process," and "improve operational accuracy." Furthermore, we gauged innovation using five items: "new products," "new services," "new channels," "new retail formats," and "new business models".

Data analysis

Given that our variables are all formatives, we followed the recommendations of Hair et al. (2019) to assess their validity and reliability. First, content validity is ensured because our variables were derived from Study 1, which had been previously validated by both practitioners and academics. This validation process ensures that each construct accurately reflects its intended concept, providing support for the achievement of content validity.

Second, we examined the formative indicators' collinearity by calculating the Variance Inflation Factor (VIF). The VIF values for all indicators were below the threshold of 3, ranging from 1.014 to 1.764, indicating no issues with collinearity – See Web Appendix E.

Third, we evaluated the statistical significance and relevance of each indicator's weight within the constructs. Items with non-significant weights and low outer loadings were reviewed for industry relevance and theoretical contribution. Consequently, we removed "Drone Delivery" from the AI in Supply Chain Management variable due to its limited adoption in retail logistics at the time of the study (Sham et al., 2023). However, we retained "AI-Powered Voice Search" (AI in Customer Service) and "AI Pricing" (AI in Marketing Management) due to their strong industry adoption and theoretical significance (Huang & Rust, 2022; Kilcourse & Rowen, 2022). To maintain the model's comprehensiveness and integrity, all other formative items were preserved, ensuring a holistic representation of AI-enabled business functions. The mean and correlation of variables are provided in Web Appendix D.

Once validated our variables, through the contrarian case analysis, we noted the existence of certain asymmetrical patterns in the database that justify the application of fsQCA (Pappas & Woodside, 2021). In addition, with the contrarian case analysis, we also stressed whether demographic variables could explain these cases (See Web Appendix H).

When conducting fsQCA, the first step involves calibrating independent and dependent variables. This process transforms each variable by assigning a new value from the calibration scale to each observation (Ragin, 2009; Rihoux & Ragin, 2008). Following established guidelines in the fsQCA literature, we employed a percentile-based direct calibration procedure using the 5th, 50th, and 95th percentiles as the thresholds for full non-membership, crossover point, and full membership, respectively. This approach is particularly suitable for datasets that lack clear theoretical thresholds or industry-standard cut-offs, which is the situation of the present study. Furthermore, the empirical distributions of our variables are right-skewed, indicating a higher concentration of firms reporting elevated levels of AI implementation, strategic emphasis, and performance. By anchoring calibration to empirical percentiles presented, we ensured that the fuzzy sets reflect meaningful variation across cases while preserving cross-case comparability, ensuring discrimination between cases, and contrasting cases near the tail of the distributions (5% and 95%). The calibration scales are presented in Table 2.

Secondly, a truth table is constructed to analyze the calibrated conditions using the algorithm embedded in the scientific package fsQCA 4.0 (Ragin & Davey, 2022).

Thirdly, thresholds for frequency and consistency are set to filter out cases that lack sufficient data for consideration in the derivation of the final configurations. The frequency threshold refers to the minimum number of observations that a row must have to be included in further analysis. In our study, this threshold is set at 1, in line with existing research practices (Douglas et al., 2020; Greckhamer et al., 2018; Pappas & Woodside, 2021). There are two types of consistency thresholds: raw consistency (a measure of how accurately a given configuration of conditions predicts the presence or absence of an outcome), set at 0.80 and PRI consistency (a prediction accuracy measure of a given configuration of conditions relative to the baseline consistency score), set at 0.70, aligning with established research practices (Douglas et al., 2020; Greckhamer et al., 2018; Pappas & Woodside, 2021). Following these steps, the next stage is generating the sufficient conditions leading to the desired outcome.

TABLE 2 Calibration scales.

	Fully-in	Crossover	Fully-out
AI in Customer Service Mgmt	6.429	5.786	4.286
AI in Store Mgmt	6.286	5.429	4.150
AI in Supply Chain Mgmt	6.421	5.429	4.143
AI in Marketing Mgmt	6.500	5.500	4.000
AI in Cybersecurity Mgmt	6.500	5.500	4.025
Efficiency	6.750	6.000	3.513
Innovation	6.800	6.200	4.400

To complement the sufficient condition analysis, we conducted a sensitivity analysis using three different calibration sets: 5%/50%/95%, 10%/50%/90%, and 20%/50%/80% (Schneider & Wagemann, 2012). Sensitivity analysis in fsQCA assesses the robustness of findings by examining how variations in calibration thresholds affect the solution consistency and coverage. The results from the first calibration set (5%/50%/95%) were consistent and are therefore reported in our findings. Detailed results of the sensitivity analysis can be found in the Web Appendix G. Additionally, we performed a necessary condition analysis on all five input variables, their negations, and the two output variables (Web Appendix F). The analysis showed that none of the variables qualify as a "must-have" condition for high organizational performance.

Findings and Discussion

The sufficient conditions for the deployment of AI leading to efficiency and innovation are presented in Tables 3 and 4, respectively. Figure 2 represents a schematic visualization of all the configurations. The analysis identified four configurations for efficiency and three for innovation. The configurations for efficiency and innovation have satisfactory levels of solution coverage (.780 and .732) and consistency (.847 and .845). In these tables, a solid checked square indicates the presence of a condition, while an empty square indicates absence. The empty cells mean the condition does not matter.

TABLE 3 Configuration deployment of AI corresponding to efficiency.

Efficiency	Solution 1	2	3	4
AI in Customer Service		✓	✓	✓
AI in Store Management		✓		✓
AI in Supply Chain Management				✓
AI in Marketing Management	✓		✓	✓
AI in Cybersecurity Mgmt	✓	✓	✓	
Raw coverage	0.6698	0.6237	0.5860	0.5453
Unique coverage	0.0943	0.0244	0.0049	0.0344
consistency	0.8753	0.8952	0.8988	0.9215
solution coverage:	0.7804			
solution consistency:	0.8472			

✓ (Indicates presence of the condition).
☐ (Indictes absence of the condition); "Empty or Blank" - It does not matter.

TABLE 4 Configuration deployment of AI corresponding to innovation.

Innovation	Solution 1	2	3
AI in Customer Service	✓	✓	✓
AI in Store Management	✓		✓
AI in Supply Chain Management		✓	✓
AI in Marketing Management			✓
AI in Cybersecurity Mgmt	✓	✓	
Raw coverage	0.6608	0.6266	0.5928
Unique coverage	0.0619	0.0278	0.0434
consistency	0.8514	0.8628	0.8992
solution coverage:	0.7319		
solution consistency:	0.8450		

✓ (Indicates presence of the condition).
☐ (Indictes absence of the condition); "Empty or Blank" - It does not matter.

The fsQCA results demonstrate that AI-enabled business functions enhance firm performance not through isolated applications but through a strategic combination across business processes. The impact of AI varies depending on how firms configure these functions to optimize efficiency, foster innovation, or achieve both.

AI configuration for efficiency

Four key AI-enabled business function configurations maximize efficiency by automating processes, improving coordination, and securing digital operations (see Table 3).

One effective configuration pairs AI marketing management and AI cybersecurity management. AI marketing automates customer segmentation and demand forecasting, while AI cybersecurity protects customer data, prevents fraud, and ensures regulatory compliance. This setup reduces manual efforts and minimizes security risks, reinforcing the balance between automation and risk mitigation (Davenport et al., 2020; Mikalef & Gupta, 2021; Teece et al., 1997; Zammuto et al., 2007).

Another configuration integrates AI customer service, AI store management, and AI cybersecurity. AI customer service optimizes workforce allocation by automating inquiries and forecasting demand. AI store management improves inventory tracking and merchandising, while AI cybersecurity safeguards transactions and prevents fraud. This integration reduces disruptions (Bankins et al., 2024; Zammuto et al., 2007) and enhances adaptability to evolving customer needs and security threats (Abadie et al., 2023; Teece et al., 1997).

A third configuration links AI customer service, AI marketing management, and AI cybersecurity management. AI customer service feeds real-time insights into AI-powered marketing, while AI cybersecurity ensures data integrity and fraud prevention. This closed-loop AI system strengthens alignment between customer interactions, marketing strategies, and security protocols (Abadie et al., 2023; Teece et al., 1997; Zammuto et al., 2007).

The most comprehensive configuration combines AI customer service, AI store management, AI supply chain management, and AI marketing management. AI customer service drives demand insights, AI store management adjusts inventory dynamically, AI supply chain management optimizes logistics, and AI marketing aligns campaigns with real-time demand forecasts. This integration synchronizes AI automation across multiple functions, allowing firms to continuously adapt and optimize business processes for sustained efficiency (Abadie et al., 2023; Teece et al., 1997).

> **Proposition 1.** *AI-enabled business functions enhance efficiency when they integrate AI-driven customer engagement, operational automation, and cybersecurity into a cohesive system that optimizes resource allocation, process coordination, and risk mitigation.*

AI configurations for innovation

Three configurations foster innovation by integrating AI-driven customer engagement, operational adaptability, and secure AI experimentation (see Table 4).

One configuration integrates AI customer service, AI store management, and AI cybersecurity management. AI customer service enables real-time personalization, while AI store management dynamically adjusts operations based on AI insights. AI cybersecurity ensures

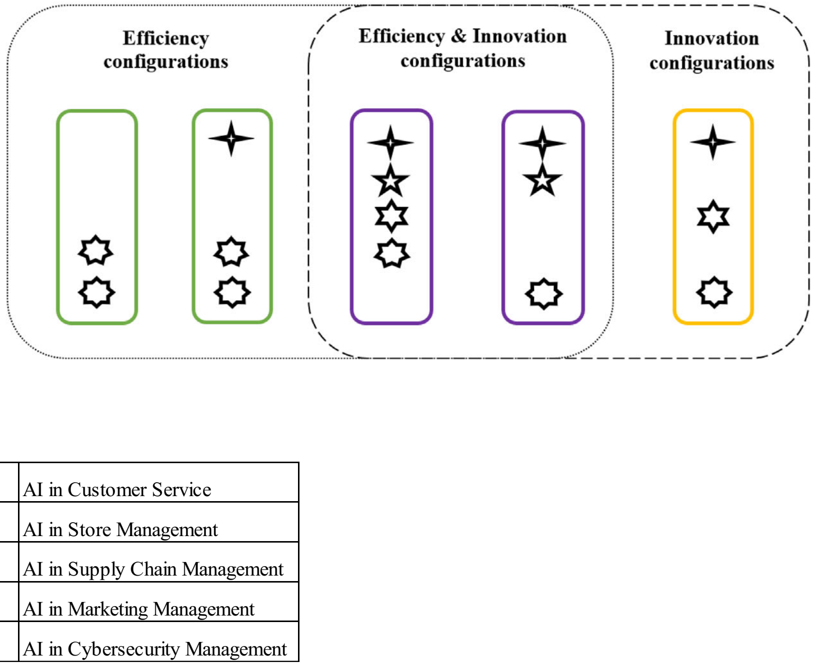

FIGURE 2 Schematic visualization of configurations.

secure data transactions, reducing risks associated with AI experimentation. This adaptive system supports firms in sensing market changes and reconfiguring processes to enhance customer experiences (Akter et al., 2023; Teece et al., 1997).

Another configuration links AI customer service, AI supply chain management, and AI cybersecurity management. AI customer service feeds demand insights into AI-powered supply chain planning, enabling firms to innovate through predictive logistics and inventory models. AI cybersecurity protects AI-driven experimentation from fraud and disruptions, ensuring that supply chain innovations remain secure. This configuration fosters AI-driven adaptability and responsiveness (Bankins et al., 2024; Zammuto et al., 2007).

A more comprehensive innovation-driven configuration includes AI customer service, AI store management, AI supply chain management, and AI marketing management. AI customer service captures evolving consumer preferences, AI store management adapts operations in real time, AI supply chain management optimizes fulfillment, and AI marketing leverages AI-driven insights for personalized engagement. This configuration supports firms in refining AI-powered business strategies and maintaining long-term adaptability (Davenport et al., 2020).

> **Proposition 2.** *AI-enabled business functions foster innovation when they create interconnected ecosystems that leverage AI-driven customer insights, operational adaptability, and secure AI experimentation.*

AI configurations for balancing efficiency and innovation

Balancing efficiency and innovation represents a persistent organizational paradox: the former demands standardization and control, while the latter requires flexibility and experimentation. Pursuing both simultaneously often leads to trade-offs that constrain firm performance (March, 1991; Raisch & Birkinshaw, 2008).

Our findings reveal two AI-enabled function configurations that consistently support both efficiency and innovation. The first combines AI in customer service, store management, and cybersecurity; the second expands to include supply chain and marketing. While structurally distinct, both configurations appear to enable capabilities that help firms reconcile the tension between process optimization and strategic renewal.

We infer three emergent capabilities – adaptive learning, predictive analytics, and uncertainty mitigation – underlying these configurations. These are not directly measured but inferred based on the functional roles of the included AI applications and supported by prior literature. Adaptive learning emerges from the real-time responsiveness of interconnected AI functions, particularly customer service and store operations. These functions facilitate iterative adjustments, enabling firms to “learn to learn” – that is, to integrate process automation for efficiency with exploratory AI applications for innovation (Božič & Dimovski, 2019).

Predictive analytics arise from the integration of data-generating functions (e.g., customer service) and execution-oriented functions (e.g., store management, supply chain, marketing). AI-enabled customer service plays a pivotal role by generating rich behavioral data through two-way interactions (Huang & Rust, 2022; Kopalle et al., 2022). When integrated with other functions, this data could be transformed into forward-looking insights that inform resource allocation, enable business model adaptation, and reduce uncertainty in experimentation (Babina et al., 2024; Verganti et al., 2020).

Uncertainty mitigation is particularly evident in the configuration that includes AI-enabled cybersecurity. This function safeguards data integrity, ensures compliance, and stabilizes AI experimentation, thereby enabling firms to pursue process optimization and innovation with reduced risk exposure (Ameye et al., 2023; Berente et al., 2021; Brock & von Wangenheim, 2019; Mikalef & Gupta, 2021). Its inclusion amplifies the effectiveness of the entire configuration by reducing the number of functions required to achieve dual outcomes.

Overall, the ability to reconcile efficiency and innovation does not stem from individual AI functions but from their coordinated deployment. This finding underscores the importance of synergistic integration – particularly involving customer service and cybersecurity – in building organization-wide AI capabilities. It marks a shift from viewing AI as a set of siloed tools toward understanding it as a dynamic configuration that enables ambidextrous outcomes (Abou-Foul et al., 2023; Mikalef & Gupta, 2021).

Proposition 3. *High-performing AI-enabled configurations reconcile efficiency and innovation by enabling adaptive learning, predictive analytics, and uncertainty mitigation through the coordinated deployment of customer service, cybersecurity (when present), and complementary AI functions.*

GENERAL DISCUSSION

Theoretical implications

This study contributes to the literature on AI in organizations by offering a multi-level framework that links AI adoption to firm performance. Rooted in a business process perspective and dynamic capability theory, our framework addresses the oversimplification observed in studies like Mishra et al. (2022), Wang et al. (2023), and Li et al. (2021), which treat AI as isolated elements impacting outcomes. Despite calls for a shift toward understanding AI's interactions with organizations (e.g., Bonetti et al., 2023; Bosse et al., 2023; Davenport et al., 2020; Huang & Rust, 2022), tackling the disorganized complexity arising from dynamic interaction remains a significant challenge. Our framework systematically explores this complexity by categorizing AI-enabled business processes at the task, business function, and firm levels.

This study advances AI integration literature by demonstrating how complementarities among AI functions drive efficiency and innovation. While prior research highlights the need for cross-functional AI coordination (Kemp, 2024; Sullivan & Fosso Wamba, 2024), it provides limited insights into how firms configure AI functions to build organization-wide AI capabilities (Akter et al., 2023; Fosso Wamba, 2022). Our findings show that AI-enabled customer service and cybersecurity serve as central enablers of emergent capabilities – adaptive learning, predictive analytics, and uncertainty mitigation – when deployed in combination with complementary functions. These capabilities reflect dynamic coordination across digital processes and help firms manage the trade-off between exploitation and exploration. In doing so, the study contributes a dynamic, process-oriented perspective on AI capability that aligns with both organizational ambidexterity and dynamic capability theory (Raisch & Birkinshaw, 2008; Teece et al., 1997). Moreover, we highlight a reciprocal relationship: while dynamic capabilities support the effective configuration of AI functions, the organization-wide AI capabilities that emerge from these configurations can, in turn, strengthen a firm's dynamic capabilities.

Our study advances the measurement of AI-enabled business functions with a context-sensitive and structured approach. Existing tools either lack contextual specificity (Abadie et al., 2023) or focus on limited aspects (Abou-Foul et al., 2023; Fosso Wamba, 2022). Building on the conceptualization of AI-enabled business functions as a set of AI-enabled tasks (Huang & Rust, 2022; Melão & Pidd, 2000), we propose a structured framework that evaluates AI implementation across three dimensions: intelligence level (augmentation vs. automation), task type, and intended objectives. Study 1 empirically validates this framework, demonstrating its effectiveness in capturing AI integration within business functions.

Managerial implications

Our findings provide actionable insights for managers: a structured approach to AI adoption is essential, allowing for assessment of implementation at the task level and ensuring that AI-driven automation and augmentation align with business function-level strategic goals. AI's impact is maximized when business functions are configured complementarily rather than in isolation. This study outlines concrete strategies to help firms achieve efficiency, innovation, or both.

To enhance efficiency, firms should adopt four key strategies. Investing in tools such as automated customer segmentation and fraud detection systems enables firms to balance demand forecasting with data security. Optimizing workforce allocation and inventory tracking while safeguarding transactions enhances operational resilience through integrated AI systems. Establishing feedback loops that use customer insights to refine marketing campaigns strengthens personalization, with cybersecurity measures ensuring data integrity. Synchronizing AI-powered customer service, store management, supply chain management, and marketing allows firms to dynamically align inventory, logistics, and advertising with real-time demand. Implementing these strategies reduces manual effort, minimizes disruptions, and ensures secure, adaptive operations.

To foster innovation, firms can leverage AI integration through three key approaches. Combining real-time personalization with dynamic operational adjustments enables agile responses to market shifts while maintaining security for AI experimentation. Linking customer insights with predictive logistics models, reinforced by AI-infused cybersecurity tools, ensures secure innovation in inventory and fulfillment strategies. Synchronizing real-time consumer data, operational agility, and personalized engagement supports AI-driven strategic refinement and long-term adaptability. These strategies empower firms to experiment safely, respond to evolving preferences, and drive AI-enabled innovation.

For firms aiming to achieve both efficiency and innovation, two strategies that integrate learning, predictive analytics, and risk mitigation apply. Securely streamlining operations through automation while leveraging customer insights fosters iterative innovation. Synchronizing customer insights with store data optimizes supply chain efficiency and enables adaptive marketing campaigns. Additionally, two business functions require particular attention in achieving this balance. Prioritizing AI in customer service enables firms to leverage automated decision-making and actionable data for hyper-personalization and market anticipation. Embedding AI in cybersecurity early in the adoption process minimizes reliance on AI integration across multiple functions while ensuring secure experimentation and operational resilience.

Successful AI integration across business functions requires anticipating and addressing key implementation challenges, including data integration issues, system interoperability constraints, and organizational resistance. Developing AI governance frameworks, establishing cross-functional AI teams, and investing in employee training are critical to ensuring smooth adoption and maximizing AI's potential for firm performance.

Limitations

Our study has several limitations, which represent opportunities for future research. First, the data in Study 1 spans 2017–2022, and Study 2 draws from U.S.-based retail firms, which may limit the generalizability of findings across time, regions, and industries. While this period captures a formative phase in AI adoption and the U.S. retail sector remains a relevant empirical setting, the recent surge in generative AI adoption post-2022 may introduce new configurations. Although our focus is on how firms structure AI-enabled functions – application patterns that typically evolve more gradually than underlying technologies – we recognize that generative AI may shift these dynamics. Future studies could extend our taxonomy to examine how emerging technologies and cross-country institutional differences reshape AI integration strategies.

Second, the results of our retail-specific study should be generalized to other sectors with parsimony, such as manufacturing or healthcare, where regulatory contexts, technological constraints, and business processes differ significantly. Comparative research across industries and geographies with varying levels of AI readiness would enhance external validity.

Third, while we infer three emergent capabilities – adaptive learning, predictive analytics, and uncertainty mitigation – from observed AI configurations and prior theory, we do not directly measure these mechanisms. In particular, the role of predictive analytics in enabling both efficiency and innovation remains theoretically grounded but empirically untested. Future research could operationalize these capabilities to assess how they mediate or moderate performance outcomes, and explore their evolution over time, especially through meta-learning and dynamic adaptation processes.

Fourth, the study is cross-sectional in nature. As such, it cannot assess the temporal stability of AI configurations or how they adapt with organizational maturity or technological change. Future research should adopt longitudinal designs by tracking AI's evolving impact on retail performance using fsQCA at three timepoints in a global sample. This builds on our study by integrating developments in generative AI (reflecting post-2022 shifts). Data on AI functions, performance, and contextual factors would be collected (e.g., strategy, regulations) at each point. FsQCA would identify transient, emergent, or persistent configurations, while dynamic capability theory interprets how firms "sense" opportunities and "reconfigure" strategies for sustained performance, addressing temporal validity. Future research can also employ a cross-cultural longitudinal design comparing configuration stability across institutional contexts with differing AI regulations (e.g., regulatory strictness), thus exploring how regulations shape AI adaptation across time. Such research could also reveal how external contexts constrain or enable the stability of the configurations across strategic cycles, organizational maturity, or external shocks, complementing the internal focus of dynamic capability theory.

Finally, our reliance on observable applications and self-reported survey data introduces limitations. Some AI deployments may remain proprietary or unreported, and executive perceptions may overstate implementation success due to social desirability or technical ambiguity. Future research could integrate alternative data sources – such as technical audits, internal reports, or digital trace data – to strengthen validity. Moreover, the cross-sectional design limits causal inference. It is possible that higher-performing firms are better positioned to invest in AI rather than AI driving performance. Longitudinal or quasi-experimental designs (e.g., difference-in-differences) would help address this endogeneity concern.

Despite these limitations, our study contributes meaningful theoretical insights and offers actionable guidance for firms navigating the complex landscape of AI-enabled business transformation.

AUTHOR CONTRIBUTIONS

Lanlan Cao: Supervision, Project Administration, Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Writing; Jiachen Yang: Methodology, Investigation, Data Curation, Formal Analysis, Writing; Yen-Tsang Chen: Data Curation, Formal Analysis, Writing.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest related to this research. There are no financial, professional, or personal relationships that could have influenced the work reported in this manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CONTRIBUTIONS TO THE GROWTH OF KNOWLEDGE

1. Introduces a multi-level framework for contextualizing AI capability through AI-enabled business processes, distinguishing task, function, and firm-level integration, and addressing the lack of contextual specificity in prior AI capability research.
2. Advances dynamic capability theory by revealing how specific configurations of AI-enabled functions – particularly involving customer service and cybersecurity – can generate organization-wide capabilities that reduce tensions between exploitation and exploration, supporting ambidextrous performance.
3. Develops and validates a replicable measurement instrument for AI-enabled business functions, grounded in qualitative coding of 6,519 real-world applications and tested via fsQCA, enabling future empirical research on the configurational effects of AI in organizational settings.

CONTRIBUTIONS TO GENERAL MANAGEMENT

1. Provides a strategic framework for managing AI across business functions, showing that coordinated deployment – rather than isolated use – drives superior performance outcomes.
2. Identifies customer service and cybersecurity as high-impact AI domains that, when integrated with other functions, unlock system-wide capabilities to balance efficiency and innovation.
3. Offers empirically grounded guidance for configuring AI portfolios, helping managers align digital initiatives with strategic priorities and design cross-functional AI strategies that support ambidextrous outcomes.

ORCID

Lanlan Cao https://orcid.org/0000-0002-4187-8333
Jiachen Yang https://orcid.org/0000-0002-3090-2465
Yen-Tsang Chen https://orcid.org/0000-0001-9057-737X

REFERENCES

Abadie, A., Roux, M., Chowdhury, S. & Dey, P. (2023) Interlinking organisational resources, AI adoption and omnichannel integration quality in Ghana's healthcare supply chain. *Journal of Business Research*, 162, 113866. Available from: https://doi.org/10.1016/j.jbusres.2023.113866

Abou-Foul, M., Ruiz-Alba, J.L. & López-Tenorio, P.J. (2023) The impact of artificial intelligence capabilities on servitization: the moderating role of absorptive capacity - a dynamic capabilities perspective. *Journal of Business Research*, 157, 113609. Available from: https://doi.org/10.1016/j.jbusres.2022.113609

Akter, S., Hossain, M.A., Sajib, S., Sultana, S., Rahman, M., Vrontis, D., et al. (2023) A framework for AI - powered service innovation capability: review and agenda for future research. *Technovation*, 125, 102768. Available from: https://doi.org/10.1016/j.technovation.2023.102768

Akter, S., Michael, K., Uddin, M.R., McCarthy, G. & Rahman, M. (2022) Transforming business using digital innovations: the application of AI, blockchain, cloud and data analytics. *Annals of Operations Research*, 308(1-2), 1–33. Available from: https://doi.org/10.1007/s10479-020-03620-w

Ameen, N., Tarhini, A., Reppel, A. & Anand, A. (2021) Customer experiences in the age of artificial intelligence. *Computers in Human Behavior*, 114, 106548. Available from: https://doi.org/10.1016/j.chb.2020.106548

Ameye, N., Bughin, J. & van Zeebroeck, N. (2023) How uncertainty shapes herding in the corporate use of artificial intelligence technology. *Technovation*, 127, 102846. Available from: https://doi.org/10.1016/j.technovation.2023.102846

Ardito, L., Filieri, R., Raguseo, E. & Vitari, C. (2025) Artificial intelligence adoption and revenue growth in European SMEs: synergies with IoT and big data analytics. *Internet Research*, 35(4), 1508–1534.

Attewell, P. & Rule, J. (1984) Computing and organizations: what we know and what we don't know. *Communications of the ACM*, 27(12), 1184–1192. Available from: https://doi.org/10.1145/2135.2136

Babina, T., Fedyk, A., He, A. & Hodson, J. (2024) Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics*, 151, 103745. Available from: https://doi.org/10.1016/j.jfineco.2023.103745

Bailey, D., Faraj, S., Hinds, P., von Krogh, G. & Leonardi, P. (2019) Special issue of organization science: emerging technologies and organizing. *Organization Science*, 30(3), 642–646. Available from: https://doi.org/10.1287/orsc.2019.1299

Balakrishnan, J. & Dwivedi, Y.K. (2024) Conversational commerce: entering the next stage of AI-powered digital assistants. *Annals of Operations Research*, 333(2), 653–687.

Bankins, S., Ocampo, A.C., Marrone, M., Restubog, S.L.D. & Woo, S.E. (2024) A multilevel review of artificial intelligence in organizations: implications for organizational behavior research and practice. *Journal of Organizational Behavior*, 45(2), 159–182. Available from: https://doi.org/10.1002/job.2735

Barley, S.R. (1988) Technology, power, and the social organization of work. *Research in the Sociology of Organizations*, 6, 33–80.

Baryannis, G., Validi, S., Dani, S. & Antoniou, G. (2019) Supply chain risk management and artificial intelligence: state of the art and future research directions. *International Journal of Production Research*, 57(7), 2179–2202. Available from: https://doi.org/10.1080/00207543.2018.1530476

Benner, M.J. & Tushman, M. (2002) Process management and technological innovation: a longitudinal study of the photography and paint industries. *Administrative Science Quarterly*, 47(4), 676–707. Available from: https://doi.org/10.2307/3094913

Berente, N., Gu, B., Recker, J. & Santhanam, R. (2021) Managing artificial intelligence. *MIS Quarterly*, 45(3), 1433–1450. Available from: https://doi.org/10.25300/MISQ/2021/16274

Bonetti, F., Montecchi, M., Plangger, K. & Schau, H.J. (2023) Practice co-evolution: collaboratively embedding artificial intelligence in retail practices. *Journal of the Academy of Marketing Science*, 51(4), 867–888. Available from: https://doi.org/10.1007/s11747-022-00896-1

Bosse, D., Thompson, S. & Ekman, P. (2023) In consilium apparatus: artificial intelligence, stakeholder reciprocity, and firm performance. *Journal of Business Research*, 155, 113402. Available from: https://doi.org/10.1016/j.jbusres.2022.113402

Božič, K. & Dimovski, V. (2019) Business intelligence and analytics for value creation: the role of absorptive capacity. *International Journal of Information Management*, 46, 93–103. Available from: https://doi.org/10.1016/j.ijinfomgt.2018.11.020

Brock, J.K.-U. & von Wangenheim, F. (2019) Demystifying AI: what digital transformation leaders can teach you about realistic artificial intelligence. *California Management Review*, 61(4), 110–134. Available from: https://doi.org/10.1177/1536504219865226

Cao, L. (2021) Artificial intelligence in retail: applications and value creation logics. *International Journal of Retail & Distribution Management*, 49(7), 958–976.

Chui, M., Hall, B., Mayhew, H., Singla, A. & Sukharevsky, A. (2022) *The state of AI in 2022*. New York, NY: McKinsey and Company.

Chui, M. & Malhotra, S. (2018) *AI adoption advances, but foundational barriers remain*. New York, NY: McKinsey and Company.

Corbin, J. & Strauss, A. (2008) *Basics of qualitative research: techniques to developing grounded theory*, 3rd edition. Los Angeles, CA: Sage.

Davenport, T., Guha, A., Grewal, D. & Bressgott, T. (2020) How artificial intelligence will change the future of marketing. *Journal of the Academy of Marketing Science*, 48(1), 24–42. Available from: https://doi.org/10.1007/s11747-019-00696-0

Dewett, T. & Jones, G.R. (2001) The role of information technology in the organization: a review, model, and assessment. *Journal of Management*, 27(3), 313–346. Available from: https://doi.org/10.1177/014920630102700306

Douglas, E.J., Shepherd, D.A. & Prentice, C. (2020) Using fuzzy - set qualitative comparative analysis for a finer - grained understanding of entrepreneurship. *Journal of Business Venturing*, 35(1), 105970. Available from: https://doi.org/10.1016/j.jbusvent.2019.105970

Fainshmidt, S., Witt, M.A., Aguilera, R.V. & Verbeke, A. (2020) The contributions of qualitative comparative analysis (QCA) to international business research. *Journal of International Business Studies*, 51(4), 455–466. Available from: https://doi.org/10.1057/s41267-020-00313-1

Fiss, P.C. (2007) A set - theoretic approach to organizational configurations. *Academy of Management Review*, 32(4), 1180–1198. Available from: https://doi.org/10.5465/amr.2007.26586092

Fosso Wamba, S. (2022) Impact of artificial intelligence assimilation on firm performance: the mediating effects of organizational agility and customer agility. *International Journal of Information Management*, 67, 102544. Available from: https://doi.org/10.1016/j.ijinfomgt.2022.102544

Fosso Wamba, S., Queiroz, M.M., Pappas, I.O. & Sullivan, Y. (2024) Artificial intelligence capability and firm performance: a sustainable development perspective by the mediating role of data-driven culture. *Information Systems Frontiers*, 26(6), 2189–2203.

Garbuio, M. & Lin, N. (2019) Artificial intelligence as a growth engine for health care startups: emerging business models. *California Management Review*, 61(2), 59–83. Available from: https://doi.org/10.1177/0008125618811931

Garvey, A.M., Kim, T. & Duhachek, A. (2023) Bad news? Send an AI. Good news? Send a human. *Journal of Marketing*, 87(1), 10–25. Available from: https://doi.org/10.1177/00222429211066972

Greckhamer, T., Furnari, S., Fiss, P.C. & Aguilera, R.V. (2018) Studying configurations with qualitative comparative analysis: best practices in strategy and organization research. *Strategic Organization*, 16(4), 482–495. Available from: https://doi.org/10.1177/1476127018786487

Guha, A., Grewal, D., Kopalle, P.K., Haenlein, M., Schneider, M.J., Jung, H., et al. (2021) How artificial intelligence will affect the future of retailing. *Journal of Retailing*, 97(1), 28–41. Available from: https://doi.org/10.1016/j.jretai.2021.01.005

Guha, S., Kettinger, W.J. & Teng, J.T.C. (1993) Business process reengineering: building a comprehensive methodology. *Information Systems Management*, 10(3), 13–22. Available from: https://doi.org/10.1080/10580539308906939

Hair, J.F., Risher, J.J., Sarstedt, M. & Ringle, C.M. (2019) When to use and how to report the results of PLS-SEM. *European Business Review*, 31(1), 2–24. Available from: https://doi.org/10.1108/EBR-11-2018-0203

Heller, J., Chylinski, M., de Ruyter, K., Keeling, D.I., Hilken, T. & Mahr, D. (2021) Tangible service automation: decomposing the technology - enabled engagement process (TEEP) for augmented reality. *Journal of Service Research*, 24(1), 84–103. Available from: https://doi.org/10.1177/1094670520933692

Hollebeek, L.D., Sprott, D.E. & Brady, M.K. (2021) Rise of the machines? Customer engagement in automated service interactions. *Journal of Service Research*SAGE Publications Sage CA: Los Angeles, CA, 24(1), 3–8. Available from: https://doi.org/10.1177/1094670520975110

Huang, M.-H. & Rust, R.T. (2022) A framework for collaborative artificial intelligence in marketing. *Journal of Retailing*, 98(2), 209–223. Available from: https://doi.org/10.1016/j.jretai.2021.03.001

Huber, G.P. (1990) A theory of the effects of advanced information technologies on organizational design, intelligence, and decision making. *Academy of Management Review*, 15(1), 47–71. Available from: https://doi.org/10.2307/258105

Kemp, A. (2024) Competitive advantage through artificial intelligence: toward a theory of situated AI. *Academy of Management Review*, 49(3), 618–635. Available from: https://doi.org/10.5465/amr.2020.0205

Kilcourse, B. & Rowen, S. (2022) *Artificial intelligence in retail: what now?*. Miami, FL: Retail Systems Research, p. 38. https://www.sas.com/content/dam/SAS/documents/analyst-reports-papers/en/rsr-artificial-intelligence-in-retail-113240.pdf

Kopalle, P.K., Gangwar, M., Kaplan, A., Ramachandran, D., Reinartz, W. & Rindfleisch, A. (2022) Examining artificial intelligence (AI) technologies in marketing via a global lens: current trends and future research opportunities. *International Journal of Research in Marketing*, 39(2), 522–540. Available from: https://doi.org/10.1016/j.ijresmar.2021.11.002

Kumar, V., Rajan, B., Venkatesan, R. & Lecinski, J. (2019) Understanding the role of artificial intelligence in personalized engagement marketing. *California Management Review*, 61(4), 135–155. Available from: https://doi.org/10.1177/0008125619859317

Li, G., Li, N. & Sethi, S.P. (2021) Does CSR reduce idiosyncratic risk? Roles of operational efficiency and AI innovation. *Production and Operations Management*, 30(7), 2027–2045. Available from: https://doi.org/10.1111/poms.13483

Mahroof, K. (2019) A human - centric perspective exploring the readiness towards smart warehousing: the case of a large retail distribution warehouse. *International Journal of Information Management*, 45, 176–190. Available from: https://doi.org/10.1016/j.ijinfomgt.2018.11.008

Manis, K.T. & Madhavaram, S. (2023) AI - enabled marketing capabilities and the hierarchy of capabilities: conceptualization, proposition development, and research avenues. *Journal of Business Research*, 157, 113485. Available from: https://doi.org/10.1016/j.jbusres.2022.113485

March, J.G. (1991) Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87. Available from: https://doi.org/10.1287/orsc.2.1.71

Melão, N. & Pidd, M. (2000) A conceptual framework for understanding business processes and business process modelling. *Information Systems Journal*, 10(2), 105–129. Available from: https://doi.org/10.1046/j.1365-2575.2000.00075.x

Meyer, A.D., Gaba, V. & Colwell, K.A. (2005) Organizing far from equilibrium: nonlinear change in organizational fields. *Organization Science*, 16(5), 456–473. Available from: https://doi.org/10.1287/orsc.1050.0135

Meyer, A.D., Tsui, A.S. & Hinings, C.R. (1993) Configurational approaches to organizational analysis. *Academy of Management Journal*, 36(6), 1175–1195. Available from: https://doi.org/10.2307/256809

Mikalef, P. & Gupta, M. (2021) Artificial intelligence capability: conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. *Information & Management*, 58(3), 103434. Available from: https://doi.org/10.1016/j.im.2021.103434

Mishra, S., Ewing, M.T. & Cooper, H.B. (2022) Artificial intelligence focus and firm performance. *Journal of the Academy of Marketing Science*, 50(6), 1176–1197. Available from: https://doi.org/10.1007/s11747-022-00876-5

Mohr, L.B. (1982) *Explaining organizational behavior*. San Francisco, CA: Jossey-Bass.

Moncrief, W.C., III. (1986) Selling activity and sales position taxonomies for industrial salesforces. *Journal of Marketing Research*, 23(3), 261–270. Available from: https://doi.org/10.1177/002224378602300306

Orlikowski, W.J. & Scott, S.V. (2008) 10 sociomateriality: challenging the separation of technology, work and organization. *Academy of Management Annals*, 2(1), 433–474. Available from: https://doi.org/10.5465/19416520802211644

Pappas, I.O. & Woodside, A.G. (2021) Fuzzy - set qualitative comparative analysis (fsQCA): guidelines for research practice in information systems and marketing. *International Journal of Information Management*, 58, 102310. Available from: https://doi.org/10.1016/j.ijinfomgt.2021.102310

Park, Y.K. & Mithas, S. (2020) Organized complexity of digital business strategy: a configurational perspective. *MIS Quarterly*, 44(1), 85–128. Available from: https://doi.org/10.25300/MISQ/2020/14477

Petrescu, M., Krishen, A.S., Gironda, J.T. & Fergurson, J.R. (2024) Exploring AI technology and consumer behavior in retail interactions. *Journal of Consumer Behavior*, 23(6), 3132–3151. Available from: https://doi.org/10.1002/cb.2386

Ragin, C.C. (2006) Set relations in social research: evaluating their consistency and coverage. *Political Analysis*, 14(3), 291–310. Available from: https://doi.org/10.1093/pan/mpj019

Ragin, C.C. (2009) *Redesigning social inquiry: fuzzy sets and beyond*. Chicago, IL: University of Chicago Press.

Ragin, C. & Davey, S. (2022) Fs/QCA. Software. Available at: https://sites.socsci.uci.edu/~cragin/fsQCA/software.shtml

Raisch, S. & Birkinshaw, J. (2008) Organizational ambidexterity: antecedents, outcomes, and moderators. *Journal of Management*, 34(3), 375–409. Available from: https://doi.org/10.1177/0149206308316058

Raisch, S. & Krakowski, S. (2021) Artificial intelligence and management: the automation–augmentation paradox. *Academy of Management Review*, 46(1), 192–210. Available from: https://doi.org/10.5465/amr.2018.0072

Rammer, C., Fernandez, G. & Czarnitzki, D. (2022) Artificial intelligence and industrial innovation: evidence from German firm-level data. *Research Policy*, 51(7), 104555. Available from: https://doi.org/10.1016/j.respol.2022.104555

Rihoux, B. & Ragin, C.C. (Eds.) (2008) *Configurational comparative methods: qualitative comparative analysis (QCA) and related techniques*. Thousand Oaks, CA: SAGE Publications.

Roberts, K.H. & Grabowski, M. (1996) Organizations, technology and structuring. In: *Handbook of organization studies*. Thousand Oaks, CA: SAGE Publications, pp. 409–423.

Rust, R.T. & Cooil, B. (1994) Reliability measures for qualitative data: theory and implications. *Journal of Marketing Research*, 31(1), 1–14. Available from: https://doi.org/10.1177/002224379403100101

Schneider, C.Q. & Wagemann, C. (2012) *Set-theoretic methods for the social sciences: a guide to qualitative comparative analysis*. Cambridge, UK: Cambridge University Press.

Sham, R., Chong, H.X., Aw, E.C.X., Thangal, T.B.T. & binti Abdamia, N. (2023) Switching up the delivery game: understanding switching intention to retail drone delivery services. *Journal of Retailing and Consumer Services*, 75, 103478. Available from: https://doi.org/10.1016/j.jretconser.2023.103478

Shankar, V. (2018) How artificial intelligence (AI) is reshaping retailing. *Journal of Retailing*, 94(4), vi–xi. Available from: https://doi.org/10.1016/S0022-4359(18)30076-9

Sorescu, A., Frambach, R.T., Singh, J., Rangaswamy, A. & Bridges, C. (2011) Innovations in retail business models. *Journal of Retailing*, 87(S1), 3–16. Available from: https://doi.org/10.1016/j.jretai.2011.04.005

Strnadl, C.F. (2006) Aligning business and it: the process - driven architecture model. *Information Systems Management*, 23(4), 67–77. Available from: https://doi.org/10.1201/1078.10580530/46352.23.4.20060901/95115.9

Sullivan, Y. & Fosso Wamba, S. (2024) Artificial intelligence and adaptive response to market changes: a strategy to enhance firm performance and innovation. *Journal of Business Research*, 174, 114500. Available from: https://doi.org/10.1016/j.jbusres.2024.114500

Teece, D.J., Pisano, G. & Shuen, A. (1997) Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7), 509–533. Available from: https://doi.org/10.1002/(SICI)1097-0266(199708)18:7<509::AID-SMJ882>3.0.CO;2-Z

Thormundsson, B. (2024) *Global AI market size 2030*. Hamburg, Germany: Statista. Available from: https://www.statista.com/forecasts/1474143/global-ai-market-size?__sso_cookie_checker=failed

Trkman, P. (2010) The critical success factors of business process management. *International Journal of Information Management*, 30(2), 125–134. Available from: https://doi.org/10.1016/j.ijinfomgt.2009.07.003

Verganti, R., Vendraminelli, L. & Iansiti, M. (2020) Innovation and design in the age of artificial intelligence. *Journal of Product Innovation Management*, 37(3), 212–227. Available from: https://doi.org/10.1111/jpim.12523

Wang, L., Zhou, Y. & Chiao, B. (2023) Robots and firm innovation: evidence from Chinese manufacturing. *Journal of Business Research*, 162, 113878. Available from: https://doi.org/10.1016/j.jbusres.2023.113878

Zammuto, R.F., Griffith, T.L., Majchrzak, A., Dougherty, D.J. & Faraj, S. (2007) Information technology and the changing fabric of organization. *Organization Science*, 18(5), 749–762. Available from: https://doi.org/10.1287/orsc.1070.0307

Zirar, A., Ali, S.I. & Islam, N. (2023) Worker and workplace artificial intelligence (AI) coexistence: emerging themes and research agenda. *Technovation*, 124, 102747. Available from: https://doi.org/10.1016/j.technovation.2023.102747

Zott, C. & Amit, R. (2007) Business model design and the performance of entrepreneurial firms. *Organization Science*, 18(2), 181–199. Available from: https://doi.org/10.1287/orsc.1060.0232

AUTHOR BIOGRAPHIES

Dr. Lanlan Cao is a Full Professor of Marketing at NEOMA BS in France. Her research interests include Digital Retailing, Retail Innovation, International Retailing, Retail Business Model Innovation, and Multichannel Retailing. Her works have appeared in the *Journal of Retailing*, *Journal of Interactive Marketing*, *International Journal of Electronic Commerce*, *International Business Review*, *Journal of Business Research*, *International Journal of Retail & Distribution Management*, *Journal of Strategic Marketing*, and other journals.

Dr. Jiachen Yang holds Ph.D. in Management from HEC Paris. He is an Assistant Professor of Strategy at NEOMA Business School. He is passionate about research on the constructive and destructive sides of digital technologies on society and business. His research appears in the *Journal of Management Studies*, *European Management Review*, *International Studies of Management and Organization*, and *Advancement in Mergers and Acquisitions*.

Dr. Yen-Tsang Chen is an Assistant Professor in the Supply Chain department at NEOMA Business School. His research interests include Behavioral Operations, Supply Chain Resilience, Operations Strategy, and the Marketing/Operations Interface. His work has been published in leading journals such as *the International Journal of Operations & Production Management*, *International Journal of Production Economics*, and *Journal of Cleaner Production*.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Cao, L., Yang, J. & Chen, Y.-T. (2025) Managing artificial intelligence across functions for enhanced retail firm performance. *European Management Review*, 1–18. https://doi.org/10.1111/emre.70042